Oliver Aleksander Larsen

Software Engineer & Researcher

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PROFILE

Software engineer and researcher who seeks out hard problems and takes them on with high ownership, focus and discipline. I build full-stack and AI-driven systems end to end, from mobile and web apps to backends, edge devices and LLM pipelines, and research how software architecture adapts when AI becomes both a design tool and a design subject. Former co-founder of a sports-nutrition startup and an endurance athlete (triathlon, running). I work with humility and curiosity, question my own assumptions, and value learning from others. I have a stutter and actively work on my speech, which has made me a more deliberate and clear communicator.

Research interests: software architecture for AI-based systems; edge computing; generative stakeholder reporting; AI governance; blockchain architecture; LLM middleware; enterprise agent substrates; decentralized agent identity; empirical studies of AI-assisted development.

EXPERIENCE

Software Architecture Researcher · Research Assistant, SDU 10/2024 – present

- Research on how software architecture adapts when AI is both a design tool and a design subject: edge computing, AI governance, LLM middleware, enterprise agent substrates, decentralized agent identity, and empirical studies of AI-assisted development. Supervised by Mahyar T. Moghaddam.
- Flood detection (2024–2025): built camera and sensor fusion on distributed edge devices (Raspberry Pi + Jetson) with a runtime adaptation manager for ML inference.
- First author on all 9 papers: 8 accepted in 2026 at IEEE ICSA, Euromicro SEAA, CASA (ECSA) and AGENTICS (IJCCI), plus 1 on SSRN.

Co-founder & Software Engineer · CarboPlanner, Odense (hybrid) 10/2024 – 2026

- Co-led full-stack development of a sports-nutrition platform for football clubs: React coach web module, Flutter mobile apps (iOS/Android), NestJS backend with MongoDB, enabling individualized meal plans, macronutrient recommendations, role-based access, and athlete and staff management.
- Focused on the backend: nutrition recommendation algorithms, agentic AI, and the end-to-end pipeline for individualized meal plans and macronutrient guidance.
- Helped raise 1,090,000 DKK in funding and supported live pilots with elite football clubs.

Teaching Assistant · University of Southern Denmark, Odense 09/2023 – present

- Teaching in Object-Oriented Programming, Data Management, Computer Systems and Software Engineering; delivered lectures for 500+ university students.

Soldier · The Royal Life Guards, Copenhagen 08/2021 – 03/2022

- Military service: 4 months of conscription and 4 months of guard duty for the Royal Danish Family.

EDUCATION & CERTIFICATIONS

MSc Software Engineering · University of Southern Denmark (SDU) 08/2025 – 06/2027 (expected)

- Supervised by Mahyar T. Moghaddam.

BSc Engineering, Software Engineering · SDU, Odense 09/2022 – 06/2025

- Grade average ~9.5 (A). Thesis on LLMs as recommender systems, grade 12 (A+).

High School Diploma · Slagelse Gymnasium, Slagelse 08/2018 – 06/2021

- Specialized in Math, Physics and Chemistry; grade average ~9 (A−). Thesis in Math and Informatics on Enigma cryptography, grade 12 (A+).

Certifications · Certified Personal Trainer · Certified Nutrition Coach

PUBLICATIONS

First author on all 9 papers: 8 accepted in 2026 (IEEE ICSA ×4, SEAA, CASA, AGENTICS ×2) + 1 on SSRN

[1] **Edge-Based Standing-Water Detection via FSM-Guided Tiering and Multi-Model Consensus. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *IEEE ICSA 2026, Software Architecture in Practice Track.*

[2] **RAD-AI: Rethinking Architecture Documentation for AI-Augmented Ecosystems. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *ANGE 2026 Workshop at IEEE ICSA 2026.*

[3] **SAGAI-MID: A Generative AI-Driven Middleware for Dynamic Runtime Interoperability. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *SAGAI 2026 Workshop at IEEE ICSA 2026.*

[4] **BitSov: A Composable Bitcoin-Native Architecture for Sovereign Internet Infrastructure. Oliver Aleksander Larsen**, Rasmus Thorsen Larsen, Mahyar T. Moghaddam. *BlockArch 2026 Workshop at IEEE ICSA 2026.*

[5] **Mining Architectural Quality Under Agentic AI Adoption: A Causal Study of Java Repositories. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *Euromicro SEAA 2026, STREAM Track.*

[6] **Persona-as-Configuration: Generative Stakeholder Reporting for Agricultural Floods. Oliver Aleksander Larsen**, Tiziano Santilli, Francesco Daghero, Mahyar T. Moghaddam. *CASA 2026 Workshop at ECSA 2026.*

[7] **NostrAgent: A Decentralized Identity and Delegation Architecture for Sovereign Agentic Systems. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *AGENTICS 2026 (IJCCI), AIPAA special session, Full Paper.*

[8] **The Agentic Company OS: Substrate Inversion for Sustained Enterprise Agent Deployment. Oliver Aleksander Larsen**, Mahyar T. Moghaddam. *AGENTICS 2026 (IJCCI), Position-paper track.*

[9] **Hybrid Large-Language-Model Meal Recommender System for Athletes: A Short Communication. Oliver Aleksander Larsen**, Vinay Chakravarthi Gogineni. *SSRN.*

PROJECTS

Hybrid LLM-Based Meal Recommender System · Bachelor's thesis 2025

- Hybrid meal recommender for CarboPlanner integrating in-context learning with LoRA/PEFT fine-tuning of Mistral-7B-Instruct, with in-context-learning and PEFT dual feedback loops for short- and long-term personalization.
- Implemented as a Flask microservice connected to the CarboPlanner Flutter app, NestJS backend and MongoDB. Achieved 46.7% accuracy and 100% format adherence on 1,000 test prompts, outperforming baselines of 41.5% without fine-tuning and 20% random. Published on SSRN.

Flood Detection System · Research project 2024 – 2025

- Real-time flood detection architecture for off-road vehicles on distributed edge devices. A Raspberry Pi captured imagery and sensor data (temperature, humidity, pressure), while YOLOv8 inference and sensor processing were routed via MQTT to either a Pi or a Jetson AGX Orin.
- A finite-state-machine runtime adaptation manager switched YOLO model sizes and forwarded processing tasks to the Pi or Jetson to balance accuracy, latency and energy. Multimodal fusion combined image predictions with sensor anomalies, using adaptive baselines for drift handling.

Stock Analysis System · Personal project 2025

- Online stock analysis hub applying logic from multi-agent LLM trading research, extended with my own implementation for improved execution. Ingests market data, runs LLM analyses, and outputs explainable reports with buy, sell and hold targets and risk-aware stop-loss levels.
- Next.js frontend with analysis cards; Flask API with MongoDB for pipelines and persistence; Celery and Upstash Redis queue async jobs to make efficient use of limited free-hosting compute.

Spark Romance Relationship App · Startup project 2023 – 2024

- Mobile app helping couples strengthen their connection through thoughtful actions and shared activities. Microservice-inspired architecture with four Flask-based APIs, a Flutter frontend and a PostgreSQL database, designed for scalability and modularity.

Enigma Simulator · High school final project (SRP) 2021

- Fully working Enigma machine simulator built in the drag-and-drop language Scratch, replicating rotor wiring, stepping and plugboard logic. Deliberately over-engineered relative to the language, an early exercise in complex system design with limited tools.

SKILLS

AI / ML	LLM integration, fine-tuning (LoRA/PEFT), in-context learning & prompt engineering, multi-agent systems, machine learning, YOLO/computer vision
Architecture	Software & database architecture, distributed edge ML (Raspberry Pi, Jetson AGX Orin), microservices, MQTT, runtime adaptation, sensor fusion
Full-stack	React, Next.js, Flutter, NestJS, Flask, MongoDB, PostgreSQL, Celery, Redis

ACHIEVEMENTS & SPORT

Ironman Copenhagen 140.6 · Triathlon: completed 3.8 km swim, 180 km bike, 42.2 km run. 08/2025

Coach Training, Stammer Program · International Program: coach in an international mentoring program supporting people with speech disabilities. 2020 – present